

Types of Routing in Core MVC & Web API (SDK 8)

List of Routing Types

1. **Conventional Routing**
 2. **Attribute Routing**
 3. **Custom Routing**
 4. **Minimal API Routing** (New in ASP.NET Core 8)
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Details with Examples and Outputs

1. Conventional Routing

Conventional routing maps incoming requests to controllers using predefined patterns. The routes are usually defined in the `Program.cs`.

Example: Route Definition (Program.cs):

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();

app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Home}/{action=Index}/{id?}");

app.Run();
```

Controller Action:

```
public class HomeController : Controller
{
    public IActionResult Index()
    {
        return Content("Welcome to the Home Page!");
    }

    public IActionResult Details(int id)
    {
        return Content($"Details of ID: {id}");
    }
}
```

Possible URLs and Outputs:

- **URL:** /
Output: Welcome to the Home Page!
(Maps to `HomeController.Index` by default)
 - **URL:** /Home/Details/5
Output: Details of ID: 5
(Maps to `HomeController.Details`)
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2. Attribute Routing

Attribute routing allows you to define routing rules directly on controller actions using attributes.

Example: Controller with Attribute Routes:

```
[Route("api/[controller]")]
public class ProductsController : ControllerBase
{
    [HttpGet]
    public IActionResult GetAll()
    {
        return Ok(new[] { "Product1", "Product2", "Product3" });
    }

    [HttpGet("{id}")]
    public IActionResult GetById(int id)
    {
        return Ok($"Product with ID: {id}");
    }
}
```

Possible URLs and Outputs:

- **URL:** /api/products
Output: ["Product1", "Product2", "Product3"]
 - **URL:** /api/products/7
Output: "Product with ID: 7"
-

3. Custom Routing

Custom routing is used to create advanced or dynamic routing scenarios. This can include constraints, areas, or multi-segment routes.

Example with Constraints:

Route Definition:

```
app.MapControllerRoute(
    name: "storeRoute",
    pattern: "store/{category}/{id:int?}",
    defaults: new { controller = "Store", action = "Browse" });
```

Controller Action:

```
public class StoreController : Controller
{
    public IActionResult Browse(string category, int? id)
    {
        return Content($"Category: {category}, ID: {(id.HasValue ?
id.Value.ToString() : "No ID")}");
    }
}
```

Possible URLs and Outputs:

- **URL:** /store/electronics
Output: Category: electronics, ID: No ID
 - **URL:** /store/books/15
Output: Category: books, ID: 15
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4. Minimal API Routing (New in ASP.NET Core 8)

Minimal APIs provide lightweight routing definitions directly in the `Program.cs` file. These are ideal for Web API scenarios.

Example: Route Definition (Program.cs):

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();

app.MapGet("/", () => "Welcome to the Minimal API!");
app.MapGet("/api/products", () => new[] { "Product1", "Product2",
"Product3" });
app.MapGet("/api/products/{id:int}", (int id) => $"Product with ID:
{id}");

app.Run();
```

Possible URLs and Outputs:

- URL: /
Output: Welcome to the Minimal API!
 - URL: /api/products
Output: ["Product1", "Product2", "Product3"]
 - URL: /api/products/7
Output: "Product with ID: 7"
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Differences in Core MVC vs Web API Routing

MVC Routing:

- Combines **Conventional** and **Attribute Routing**.

Example:

```
app.MapControllerRoute(  
    name: "default",  
    pattern: "{controller=Home}/{action=Index}/{id?}");
```

Web API Routing:

- Primarily **Attribute Routing** or **Minimal APIs**.

Example:

```
app.MapGet("/api/items", () => "Items");
```

Complete Comparison Table

URL	Routing Type	Controller Action/Handler	Output
/	Conventional	HomeController.Index()	Welcome to the Home Page!
/Home/Details/5	Conventional	HomeController.Details(5)	Details of ID: 5
/api/products	Attribute/Minimal	ProductsController.GetAll()	["Product1", "Product2", "Product3"]
/store/books/15	Custom with Constraints	StoreController.Browse()	Category: books, ID: 15
/api/products/7	Minimal API	Lambda Route Handler	Product with ID: 7